12. List and Keys in React

▼ What is a List in React?

- A **list** is a collection of elements that you render **dynamically** from an array.
- We **loop through arrays** in JSX using **JavaScript's** `.map()` **method** to render each item as a React element.

▼ Why Use Lists in React?

Lists help to:

- Avoid writing repeated HTML
- Display dynamic data
- To **render multiple similar components**

▼ Example: Rendering a Simple List

- **Code:**

```
import React from "react";

const NamesList = () => {
  const names = ["John", "Bob", "Smith", "Doe", "Dave"];
  return (
    <div>
      <h2>Name List:</h2>
      {names.map((name, index) => {
        return <p>{name}</p>;
      })}
    </div>
  );
};

export default NamesList;
```

- **Output:**

Name List:

John

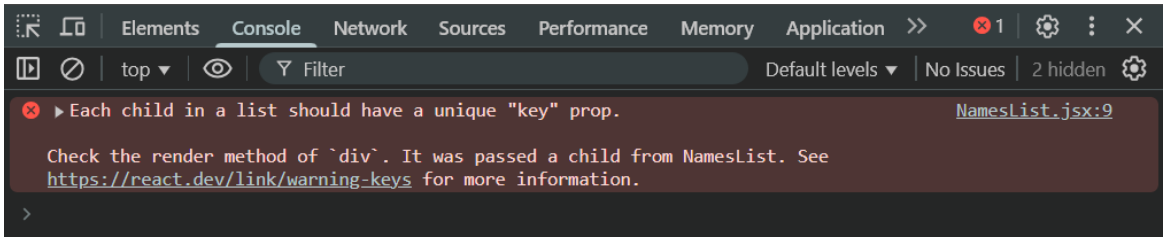
Bob

Smith

Doe

Dave

- You will get the desired output but along with that one error in the console.



- So basically, React does not identify which element has been added, removed or updated. To track the elements uniquely, we have to pass the unique ID to each and every element. Here unique ID is the index value.
- So we need keys in React JS to provide a unique key for each and every element.

▼ What is Key in React?

- A **key** is a **special prop** in React.
- A key is used to give a unique identity to each and every element.
- **Purpose of Keys:**
  - Helps React **identify which items changed**, are added, or removed.
- **Important Note**: **A key should be unique for each list item.**
- **Updated Code:**

```
import React from "react";

const NamesList = () => {
  const names = ["John", "Bob", "Smith", "Doe", "Dave"];
  return (
    <div>
      <h2>Name List:</h2>
      {names.map((name, index) => {
        return <p key={index}>{name}</p>;
      })}
    </div>
  );
};

export default NamesList;
```

▼ Example: List of Objects

- **Code:**

```
import React from "react";

const UsersList = () => {
  let starcast = [
    { id: 101, name: "Eleven" },
    { id: 102, name: "Mike" },
    { id: 103, name: "Will" },
    { id: 104, name: "Lucas" },
    { id: 105, name: "Dustin" },
  ];
  return (
    <div>
      <h2>Stranger Things Starcast</h2>
      <ul>
        {starcast.map((character, index) => {
          return <li key={character.id}>{character.name}</li>;
        })}
```

```
      </ul>
    </div>
  );
};

export default UsersList;
```

- **Output: You can pass either the `id` or the `index` both will work.**

Stranger Things Starcast

- Eleven
- Mike
- Will
- Lucas
- Dustin

▼ Dynamically Adding/Removing Items with Keys

- **Code:**

```
import React, { useState } from "react";

const Tasks = () => {
  const [tasks, setTasks] = useState(["Task A", "Task B"]);

  const addTask = () => setTasks([...tasks, "Task C"]);

  return (
    <div>
      {tasks.map((task, i) => (
        <p key={i}>{task}</p>
      ))}
      <button onClick={addTask}>Add Task</button>
    </div>
  );
```

```
};

export default Tasks;
```

- **Output: Before adding to the task**

Task A

Task B

Add Task

- **After Adding the task**

Task A

Task B

Task C

Add Task